How do we know who won?

One subscriber wins a fictional $1 million giveaway. The announced ticket should be something any entrant can check.

CURBy-Q ROUND 28297

Binary 011 means 3. Add 1. Ticket #4 wins.

What problem does this solve?

A subscriber watching a prize announcement has to trust the creator. Our verifier lets them use the same public number and the same rule to check it themselves. Announcing #4 passes. Announcing #7 produces a mismatch.

Could ordinary Python do this?

Yes. Python can check the ticket, and a classical public beacon could also support replay. The quantum value comes from how CU Boulder produces its public number. Measurements on entangled particles give the beacon evidence of unpredictability before publication.

What did your team build?

The giveaway verifier, the ticket mapping and its tests, and a Qiskit notebook connecting the receipt to a two-player quantum game. We reused CU Boulder's public record and IBM's game. In 8,000 simulated trials, our entangled strategy won 85.56%. The best classical strategy wins 75%.

What could the organizer still manipulate?

They could omit a subscriber, rearrange the list, or choose a round after seeing its value. A real giveaway must publish the ordered tickets, selection rule and future round in advance, and let entrants check inclusion. This historical replay checks which listed ticket won.